



Department of Information Technology

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch : V Semester-B. Tech Information Technology

Course Code & Title : IT3501 Full Stack Web Development

Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 1 / Role of Express
- **Course Outcome** : CO1
- **Topic Learning Outcome:** TLO4
- **Activity Chosen** : Reflection
- **Justification:**
 - Reflection activity makes it easier to assess students grasp of a concept and uncover doubts or unclear points in their learning.
 - Express plays a key role in Node.js development by simplifying the process of building web applications and APIs.
 - This activity was chosen to help students understand the role of Express in improving Node.js development by providing easy routing, middleware handling, and a well-structured application design.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 - The activity took place at the end of the class after a detailed explanation of role of Express, including route configuration, middleware implementation, and a demonstration of an Express server as shown in Figure 1
 - Instructor explained how to create a server environment using Express backend framework.
 - Students expressed their understanding and listed any doubts they had about the lesson.
 - After submitting their responses, the instructor redistributed them in the next class as shown in Figure 2.
 - This activity helped the students to improve their understanding level.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO10	PSO1
CO3	3	2	2	2	1	3

(1 – Low

2 – Moderate

3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO5	PO10	PSO1
	3	2	2	2	1	3
Justification for correlation	Students will be able to understand the fundamental role of Express in simplifying backend development.	Students will be able to analyze routing and middleware-related problems using Express.	Students will be able to develop modern web applications using Express to solve real-world backend challenges.	Students will be able to utilize Express to handle dynamic routes, APIs, and request management efficiently.	Students will be able to communicate their understanding of Express-based server development effectively within a team.	Students will be able to apply Express's capabilities in creating scalable and maintainable backend systems.

• Images / Screenshot of the practice:



Fig 1: Reflection Activity

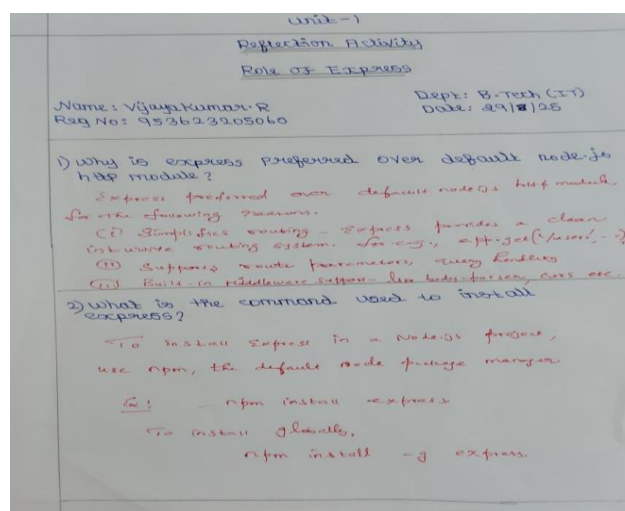


Fig 2: Sample Sheet of Reflection Activity

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- Students were motivated to self-assess and determine which steps in Express, including routing, middleware application, and handling requests and responses, required additional understanding.
- Students reported that the activity encouraged them to express their doubts more freely and interact with the instructor in a meaningful way.

❖ *Benefit of the practice:*

- The activity made it easier to identify students who struggled with the conceptual understanding of Express and its role in backend development.
- It enabled the instructor to offer targeted clarifications using example programs, ultimately enhancing student comprehension.

❖ *Challenges faced in implementation:*

- Some students struggled to understand how middleware flows in sequence and how error-handling functions work in Express.
- Few students opted not to ask any questions, thus I was unable to really get an understanding of all the students' perspectives.

In the following class, the instructor addressed all the students' questions in detail, ensuring that their doubts about the role and use of Express were completely resolved.

References:

- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/RE_C_BV_22-23_U%20III.pdf
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1.KVS_Reflection_FDS.pdf

Signature of Faculty Member

HOD